Surya Doddipatla

Machine Learning Engineer | **Computer Vision & LLM Applications** | **Azure MLOps**

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Professional Summary

Data Scientist and ML Engineer on a small team. At Dream Forge I built an AI floor-plan platform from scratch: the computer vision and deep learning models, LLM features chosen through a structured evaluation, and the Azure platform they run on (six Container Apps services, Key Vault, CI/CD, dev and prod). Before that I built Azure ETL for a US energy utility at Accenture.

Technical Skills

- **Machine Learning & Computer Vision:** PyTorch, SegFormer, image segmentation, Vision Transformers, OpenCV, OCR, scikit-learn
- **Geometry & Spatial:** Shapely, NetworkX, computational geometry, constraint-based layout, DXF and OBJ export, GIS
- **Generative AI & NLP:** LLM evaluation (GPT-4o, Gemini, Claude), speech-to-text with LLM refinement, RAG, embeddings
- **MLOps & Cloud:** Docker, Azure (Container Apps, ACR, Blob, Key Vault, Static Web Apps, Data Factory), GitHub Actions CI/CD, AWS
- **Software & Data:** Python, SQL, Flask, FastAPI, REST APIs, JavaScript, React, Next.js, Pandas, SQL Server, ETL, Databricks

Experience

Dream Forge Homes Oct 2025 – Present
Data Scientist *Maryland, USA*

- Working directly with the CEO of a small startup, designed and built a full-stack AI platform from scratch, used daily by architects to generate and present design concepts for residential remodel and rebuild projects; replaced manual drafting in early client engagement.
- Built the computer vision pipeline (OpenCV, NetworkX, Shapely, EasyOCR) that converts hand-drawn and scanned floor plans into CAD-ready wall, room and opening geometry at **98 to 100% accuracy** across residential plan sizes and layouts.
- Built and deployed a PyTorch SegFormer segmentation service for scanned plans where classical CV falls short, behind the same API so the editor switches between the two with no client changes.
- Designed a furniture placement system, built from annotated real project data, that produces deterministic layouts under clearance, adjacency and door-swing constraints with **zero API calls** at inference.
- Ran a structured LLM evaluation across leading models that set the product's architecture and removed projected ongoing API costs; shipped speech-to-text project descriptions with LLM refinement and offline-safe recording.
- Split the Flask monolith into **six** independently deployed Azure Container Apps services (Docker, ACR, Key Vault via managed identity, GitHub Actions CI/CD gated on **160+ tests**); set up dev and prod and the APIs partner teams integrate through.
- Cut production container memory **80%** through selective model loading; built DXF (CAD) and OBJ (3D) exporters, a GIS site-context layer, and project version backups with restore.

UMBC Observatory, University of Maryland Baltimore County Oct 2024 – Jan 2026
Engineer (Part-time) *Baltimore, MD*

- Built the shift-scheduling system for student telescope operators across multiple daily shifts, enforcing fairness and safety constraints (no back-to-back overnight shifts, equal early-shift share, class availability).
- Developed a Python forecasting model that predicts telescope over-utilization at **87% accuracy**, used to flag weeks needing extra coverage; automated weekly and monthly SQL Server and Azure reporting that replaced manual work.

Accenture Nov 2022 – Jul 2023
Associate Software Engineer *Pune, India*

- Developed and maintained ETL jobs on the Azure data platform of a large US energy utility (Data Factory, Data Lake Storage, Azure SQL) under senior engineers, moving operational and customer data into reporting systems and fixing daily-load failures.
- Tuned SQL queries and pipeline steps on assigned reporting workloads, cutting runtime by **65%**; helped the data science team move model outputs from notebooks into scheduled production jobs and client dashboards.

Projects

- **cited-extract** (open source, Python, github.com/Surya-dcode/cited-extract): extracts fields from any PDF with a quote and page for every value, checks each quote against its page in code, and measures how often the model invents values.

Education

University of Maryland, Baltimore County Aug 2023 – May 2025
Master of Science in Data Science, GPA 3.7 *Baltimore, MD*

Vidya Jyothi Institute of Technology (JNTU Hyderabad) Jul 2019 – May 2022
Bachelor of Engineering in Mechanical Engineering *Hyderabad, India*

Certifications

AWS Data Engineer – Associate · AWS Cloud Practitioner · Databricks Generative AI Fundamentals · Databricks Fundamentals · Microsoft PL-300